

Chapter 4: Exponential and Logarithmic Functions

Name: _____

Date: _____

Instructions: Answer all questions. Show complete work and state domain restrictions whenever relevant. Unless otherwise stated, give exact answers; for numerical approximations, round to two decimal places.

Convention: $\ln$ denotes the natural logarithm. When no base is shown, $\log$ means natural $\log$; $\log_b$ denotes base- b .

1. Let $f(x) = \ln\left(\frac{x+2}{x-1}\right)$. (12)

(a) Determine the domain and range of f .

(b) Find $f^{-1}(x)$ and state its domain.

(c) Verify by composition that f and f^{-1} are inverses on their domains.

2. Let $f(x) = 2\log_3(x^2 - 4x - 5)$. (12)

(a) Find the domain and range of f .

(b) Solve $f(x) = 2$ and give exact solutions.

(c) On which intervals of its domain is f one-to-one? Briefly justify.

3. If $c = \log_y b$ with $c \neq 0$ and $d = 2 \log_{y^5}(b^3)$, find $\frac{d}{c}$ in simplest form. Do not evaluate b or y numerically. (10)

4. Use the Laws of Logarithms to expand the expression completely and indicate all domain restrictions: (10)

$$\log \left(\sqrt{\frac{x+2}{x(x^2+4)}} \right).$$

5. Combine into a single logarithm and simplify, stating the domain restrictions: (10)

$$\ln x - 2 \ln(x^2 + 1) + \frac{1}{2} \ln(3 - x^4).$$

6. Solve each equation for real solutions. Give exact answers when possible or approximate to two decimal places. (16)

(a) $2^{x-1} = 10$

(b) $5 \ln(3 - x) = 4$ (state the domain restriction first)

(c) $10^{x+3} = 6^{2x}$

(d) $\log_3(x + 2) + \log_3(x - 1) = 2$

(e) Solve for t : $t = \ln(2 - e^t) - \ln 3$.

7. Suppose a radioactive element has a half-life of $t_{1/2}$ and initially there are N_0 atoms in a container. After time t , there are N_t atoms remaining. (14)

(a) Explain why $\left(\frac{1}{2}\right)^{t/t_{1/2}} = \frac{N_t}{N_0}$.

(b) Take natural logarithms of both sides to show that $\frac{t}{t_{1/2}} \ln\left(\frac{1}{2}\right) = \ln\left(\frac{N_t}{N_0}\right)$.

(c) Hence show that $t = -t_{1/2} \frac{\ln\left(\frac{N_t}{N_0}\right)}{\ln 2}$.

8. The initial size of a culture is unknown. After 30 minutes the count is 2500, and after 110 minutes the count is 18,000. Assume continuous exponential growth $P(t) = P_0 e^{kt}$ with t in minutes. (14)

(a) Determine P_0 and k by solving a system.

(b) Predict when the population will reach 50,000.

(c) Compute the doubling time and interpret it in context.

9. If $\frac{4^x}{2^{x+y}} = 8$ and $\frac{9^{x+y}}{3^{5y}} = 243$, where x and y are real numbers, find the ordered pair (x, y) . (10)